

Software Requirements Specification (SRS)

for

Smart E-Learning Platform (SELP)

Version 1.0

Prepared by: [Your Company Name] - Product Team

Date: October 26, 2023

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the requirements for the Smart E-Learning Platform (SELP). This document will serve as the foundation for the design, development, and testing phases. It defines the functional and non-functional requirements, user interfaces, and system constraints.

1.2 Document Conventions

- Must: Indicates a mandatory requirement.
- Should: Indicates a recommended requirement.
- May: Indicates an optional requirement.
- User: Refers to a student or educator.
- Admin: Refers to system administrator.

1.3 Intended Audience

- Development Team: To implement the system.

- Project Managers: To plan timelines and resources.
- Quality Assurance (QA) Team: To create test cases.
- Stakeholders: To validate that the system meets business needs.

1.4 Product Scope

The SELP is a web-based application designed to facilitate virtual learning. It allows educators to upload video lectures, create quizzes, and track student progress. Students can access materials, submit assignments, and interact with peers via discussion forums. The system aims to replace outdated Learning Management Systems (LMS) with an AI-driven recommendation engine that suggests content to students based on their learning pace.

1.5 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- SELP Business Requirements Document (BRD) v1.0.

2. Overall Description

2.1 User Needs

- Students: Need an intuitive interface to access courses, complete assessments, and receive feedback.
- Educators: Need tools to create/upload course materials and manage student performance.
- Admins: Need to manage users and system configurations.

2.2 Assumptions and Dependencies

Assumptions:

- Users have a stable internet connection.
- Users possess a modern web browser (Chrome, Firefox, Edge).
- Educators are proficient in creating digital content (video, PDF).

Dependencies:

- The system relies on a third-party Cloud Storage Service (e.g., AWS S3) for hosting media files.
- The AI recommendation engine depends on the availability of user historical performance data.

3. System Features and Requirements

3.1 Functional Requirements

3.1.1 User Authentication & Authorization

FR-01: The system must allow users to register using an email address and password.

FR-02: The system must support Single Sign-On (SSO) via Google and Microsoft accounts.

FR-03: The system must enforce Role-Based Access Control (RBAC): Student, Educator, and Admin.

FR-04: The system must lock user accounts after 5 unsuccessful login attempts.

3.1.2 Course Management (Educator)

FR-05: Educators must be able to create a new course by entering a Title, Description, and Category.

FR-06: Educators must be able to upload video lectures (MP4 format) and supporting PDF documents.

FR-07: Educators must be able to schedule course availability (Start Date / End Date).

FR-08: Educators should be able to organize content into Modules/Chapters.

3.1.3 Content Consumption (Student)

FR-09: Students must be able to browse the course catalog by category or search keywords.

FR-10: Students must be able to view video lectures with a "Resume" feature (remembering the last watched timestamp).

FR-11: The system must automatically mark a lecture as "Completed" once the student watches 90% of the video.

3.1.4 Assessments and Grading

FR-12: Educators must be able to create multiple-choice quizzes (MCQs).

FR-13: The system must automatically grade MCQ quizzes and provide instant feedback to the student.

FR-14: Educators must be able to create subjective assignments (upload file submission).

FR-15: Educators must be able to enter grades manually for subjective assignments.

FR-16: Students must be able to view their overall grade and progress percentage per course.

3.1.5 AI Recommendation Engine

FR-17: The system must analyze a student's quiz results and viewing speed to recommend supplementary content.

FR-18: If a student scores below 60% on a quiz, the system must recommend a "Remedial Video" on the failed topic.

3.1.6 Communication

FR-19: The system must have a discussion forum for each course where students can post questions.

FR-20: Educators must receive notifications when a new question is posted in their course forum.

FR-21: The system must notify students via email when their assignments are graded.

3.2 External Interface Requirements

3.2.1 User Interfaces

UI-01: The web interface must be responsive and adaptable to desktop, tablet, and mobile devices.

UI-02: The dashboard must display a progress bar for ongoing courses.

UI-03: The video player must have standard controls (Play, Pause, Volume, Fullscreen).

3.2.2 Hardware Interfaces

HW-01: The system must support standard webcams and microphones for live-streaming features (if applicable).

3.2.3 Software Interfaces

SI-01: The system must interface with an SMTP server (e.g., SendGrid) to send transactional emails.

SI-02: The system must interface with an S3-compatible storage bucket for media assets.

SI-03: The system must interface with an AI model API (e.g., TensorFlow Serving) for recommendation logic.

3.3 Non-Functional Requirements

3.3.1 Performance

NFR-01: The platform must support up to 1,000 concurrent users with a response time under 2 seconds.

NFR-02: Video lectures must start buffering within 3 seconds of the play button being pressed.

NFR-03: The database query time for the course catalog must average less than 500ms.

3.3.2 Security

NFR-04: All data transmitted between the client and server must be encrypted using TLS 1.3.

NFR-05: User passwords must be hashed using BCrypt.

NFR-06: User data privacy must comply with GDPR regulations (right to be forgotten, data export).

3.3.3 Availability

NFR-07: The system must maintain 99.9% uptime (excluding scheduled maintenance windows).

NFR-08: The system must perform automated backups of the database every 24 hours.

3.3.4 Usability

NFR-09: The system must be localized for English and Spanish languages.

NFR-10: 95% of new users must be able to successfully enroll in a course without assistance.

3.3.5 Reliability

NFR-11: In the event of a video upload failure, the system must display a clear error message and allow the user to retry without losing the course setup.

Appendices

- Appendix A: Database Schema Diagram (To be drafted).

- Appendix B: Wireframes (To be drafted).
- Appendix C: Glossary of Terms.